

Session 14: Kubernetes Troubleshooting

Task 1: Kubernetes Commands

Perform hands-on practice with all important troubleshooting commands.

Cover:

kubectl get

kubectl describe

kubectl logs

kubectl exec

kubectl events

kubectl explain

kubectl top

kubectl get -o wide

Task 2: Troubleshoot Common Issues

Practice troubleshooting:

- CrashLoopBackOff
- ImagePullBackOff

- ErrImagePull
- Pending
- ContainerCreating
- Service connectivity issues
- DNS issues
- Pod networking issues
- Configuration issues

For every issue:

1. Identify the problem.
2. Investigate.
3. Find the root cause.
4. Fix it.
5. Verify the solution.
6. Document the troubleshooting process.

Task 3: Mini Project

Complete the Kubernetes troubleshooting mini project.

Deliverables

- Commands
- Problem statement
- Investigation steps
- Root cause
- Solution
- Before/after output
- Screenshots
- README.md